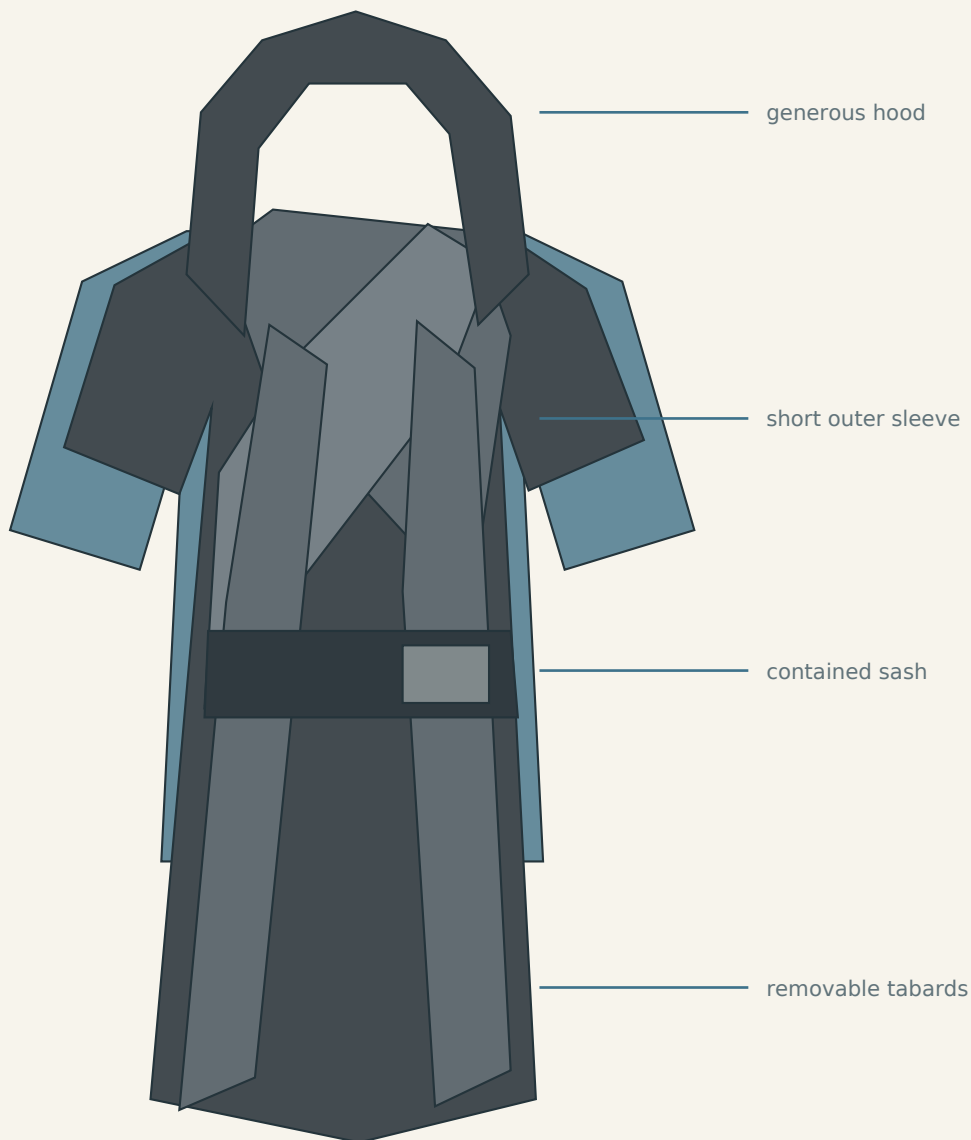


Wear the curiosity.

A practical first draft of the clothing in the Electronic Scouts concept images.



ONE PATTERN FAMILY. YOUR MEASUREMENTS.

Charcoal wrap robe, a cord-free hood, blue short-sleeve undertunic and optional tabards. Keep the movement easy and the cloth matte. The device carrier remains a separate design until its real dimensions and weight are known.

Measure the wearer.

All inputs and outputs use millimeters. Example values demonstrate the draft; they are not a standard child size.

chest Full chest over the clothes to be worn underneath.	680 mm
hip Fullest seat/hip over underlayers.	720 mm
waist Natural waist, comfortably measured.	630 mm
shoulder_width Shoulder point to shoulder point across the back.	310 mm
shoulder_to_waist High shoulder point beside neck down to waist.	340 mm
neck_circumference Around base of neck; ease is separate.	300 mm
head_circumference Around brow and widest back of head.	540 mm
head_height Vertical shoulder-level-to-crown distance; use a helper.	285 mm
upper_arm Around fullest upper arm over intended underlayer.	230 mm
hand_circumference Around hand at widest knuckles, thumb tucked.	170 mm
wrist Wrist circumference; relevant to long, narrower sleeves.	140 mm
robe_length High shoulder point to chosen robe hem.	750 mm
tunic_length High shoulder point to chosen tunic hem.	510 mm
sleeve_length Shoulder point to outer sleeve edge; NOT sleeve pattern length.	180 mm
tunic_sleeve_length Shoulder point to inner sleeve edge.	230 mm

Measure over the intended underlayers, then add ease through the separate ease controls. Do not scale every dimension by age or height: heads, hands, shoulders and torso proportions vary independently.

Set the character.

Edit measurements.example.json, save a wearer-specific copy, then regenerate the entire set.

FIT + WRAP

chest_ease	160 mm
hip_ease	160 mm
sleeve_ease	120 mm
cuff_ease	70 mm
neck_ease	40 mm
wrap_overlap	95 mm
hem_flare	35 mm

HOOD + ACCESSORIES

hood_ease	40 mm
sash_width	50 mm
sash_overlap	80 mm
sash_ease	40 mm
tabard_width	85 mm
tabard_drop	110 mm

FINISH + COMPONENTS

seam_allowance	10 mm
hem_allowance	25 mm
binding_width	10 mm
include_hood	yes
include_tunic	yes
include_tabards	yes
include_sash	yes

A draft is a hypothesis.

The formulas match adjoining seam lengths and produce a consistent cutting shape. They cannot establish comfort, hood balance, drape or fit on a particular wearer. Make a toile in inexpensive fabric with similar weight before cutting the good cloth.

Choose the first silhouette.

The example has short outer sleeves, a slightly longer blue sleeve underneath and an activity-length hem. Length, ease and accessories are independent choices. Long sleeves need a fresh reach and cuff check.

Finish the fit at the waist.

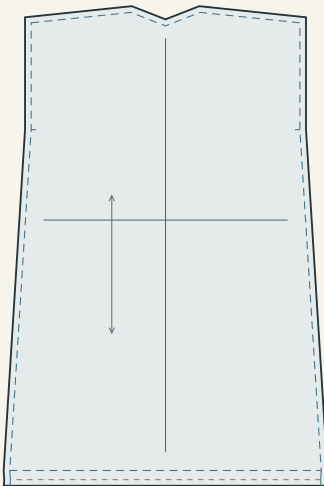
Use a broad, short fastening sash with hook-and-loop tabs. Position closures during fitting. Keep the hood cord-free; let detachable decoration provide the ceremony.

The cuts.

Black outlines include allowances. Dashed blue lines show seam or finished boundaries. Diagrams on this page are not to scale.

robe_back / Outer robe - full back

CUT 1 FULL PIECE; no fold



Cut envelope: 531.0 x 786.2 mm. Charcoal outer cloth

Trace the full piece once, not on a fold.

Join shoulders and side seams at seam allowance; stop at underarm marks.

Staystitch the angular back neckline. Clip inside the V allowance only as needed, stopping short of stitching.

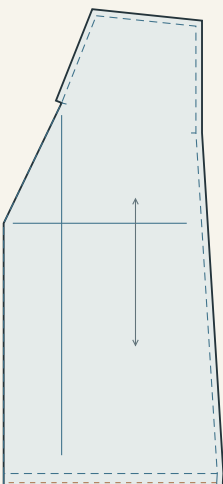
Bottom hem: turn 10 mm, then 15 mm to the marked finished line.

MARKS ON FULL-SIZE CUTS

M1 CENTRE BACK / M2 SASH LEVEL / M3 UNDERARM / M5 FIRST TURN 10 MM

robe_front / Outer robe - wrap front

CUT 2 MIRRORED (one left, one right)



Cut envelope: 361.0 x 786.7 mm. Charcoal outer cloth

Cut one left and one right; do not cut two identical handed fronts.

Front openings have zero allowance and receive double-fold bias binding.

The diagonal below the marked neck endpoint is the wrap edge; do not sew it to the hood.

Bottom hem: turn 10 mm, then 15 mm to the marked finished line.

Position short internal hook-and-loop tabs during fitting; no long internal ties are required.

MARKS ON FULL-SIZE CUTS

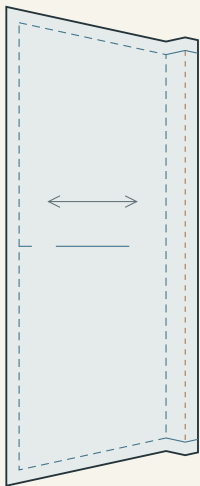
M1 CENTRE FRONT / M2 SASH LEVEL / M3 UNDERARM / M4 HOOD ENDS HERE / M5 FIRST TURN 10 MM

The cuts.

Black outlines include allowances. Dashed blue lines show seam or finished boundaries. Diagrams on this page are not to scale.

robe_sleeve / Outer robe - flat sleeve

CUT 2 FULL PIECES; no fold



Cut envelope: 151.0 x 375.8 mm. Charcoal outer cloth

The long body edge joins the front and back arm openings without gathering.

Match centre notch on attachment edge to shoulder seam; remaining ends are underarms.

Join the two equal sloping underarm edges to close each sleeve.

Cuff hem: turn 10 mm, then 15 mm. The cut extension is reflected at both folds to match the sleeve taper.

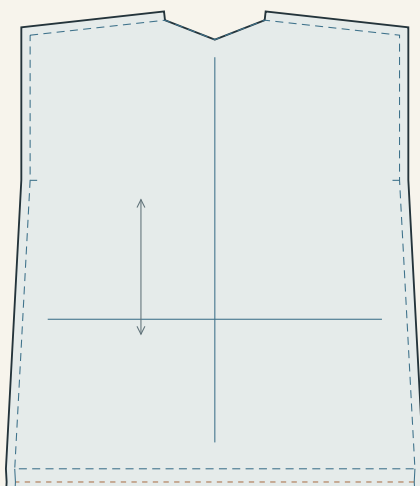
Follow the marked underarm seam extensions through the shaped hem allowance; do not trim them to a straight continuation.

MARKS ON FULL-SIZE CUTS

M1 SHOULDER SEAM / M2 TOP OF SLEEVE / M3 FIRST TURN 10 MM

tunic_back / Inner tunic - full back

CUT 1 FULL PIECE; no fold



Cut envelope: 476.0 x 545.9 mm. Slate-blue inner cloth

Trace the full piece once, not on a fold.

Join shoulders and side seams at seam allowance; stop at underarm marks.

Bind neckline raw/final edge; zero extra allowance on those two edges.

Bottom hem: turn 10 mm, then 15 mm to the marked finished line.

MARKS ON FULL-SIZE CUTS

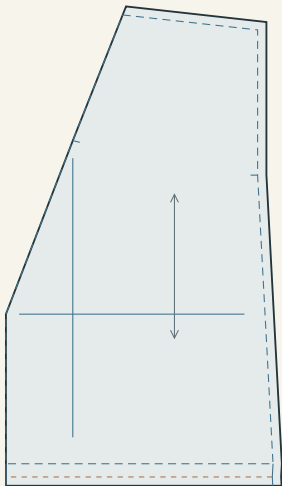
M1 CENTRE BACK / M2 SASH LEVEL / M3 UNDERARM / M5 FIRST TURN 10 MM

The cuts.

Black outlines include allowances. Dashed blue lines show seam or finished boundaries. Diagrams on this page are not to scale.

tunic_front / Inner tunic - wrap front

CUT 2 MIRRORED (one left, one right)



Cut envelope: 314.5 x 545.6 mm. Slate-blue inner cloth

Cut one left and one right; do not cut two identical handed fronts.

Front openings have zero allowance and receive double-fold bias binding.

Bind neckline and front opening as one continuous edge after joining shoulders.

Bottom hem: turn 10 mm, then 15 mm to the marked finished line.

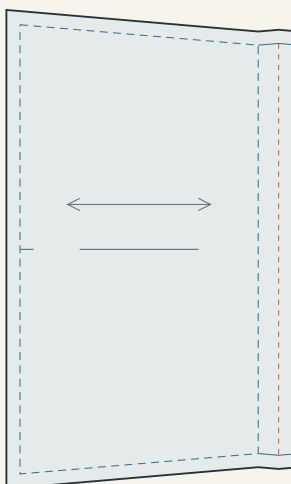
Position short internal hook-and-loop tabs during fitting; no long internal ties are required.

MARKS ON FULL-SIZE CUTS

M1 CENTRE FRONT / M2 SASH LEVEL / M3 UNDERARM / M4 NECK/WRAP CORNER / M5 FIRST TURN 10 MM

tunic_sleeve / Inner tunic - flat sleeve

CUT 2 FULL PIECES; no fold



Cut envelope: 211.0 x 352.8 mm. Slate-blue inner cloth

The long body edge joins the front and back arm openings without gathering.

Match centre notch on attachment edge to shoulder seam; remaining ends are underarms.

Join the two equal sloping underarm edges to close each sleeve.

Cuff hem: turn 10 mm, then 15 mm. The cut extension is reflected at both folds to match the sleeve taper.

Follow the marked underarm seam extensions through the shaped hem allowance; do not trim them to a straight continuation.

MARKS ON FULL-SIZE CUTS

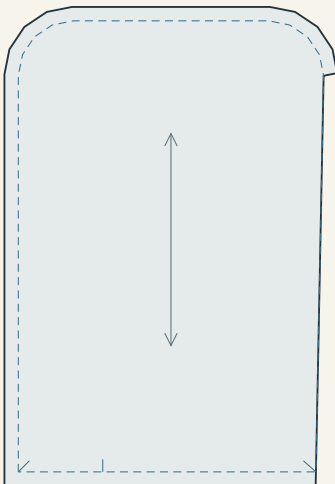
M1 SHOULDER SEAM / M2 TOP OF SLEEVE / M3 FIRST TURN 10 MM

The cuts.

Black outlines include allowances. Dashed blue lines show seam or finished boundaries. Diagrams on this page are not to scale.

robe_hood / Outer robe - hood side

CUT 2 MIRRORED (one left, one right)



Cut envelope: 240.8 x 346.0 mm. Charcoal outer cloth

Join the two panels around the back and crown; face edge remains open.

Bottom attachment edge runs from centre-back to the marked front neckline endpoint.

Match shoulder notch to body shoulder seam; hood face and garment wrap edges meet at the front neck endpoint.

Bias-bind the face opening with zero extra edge allowance. No drawcord channel.

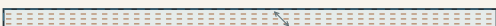
Crown is drafted as short straight segments; sew along the supplied polyline for the first toile.

MARKS ON FULL-SIZE CUTS

M1 SHOULDER / M2 CENTRE BACK / M3 FRONT NECK END

robe_bias_strip / Outer - bias binding strip

CUT 2 STRIPS ON TRUE BIAS; may piece shorter strips



Cut width 40 mm on TRUE BIAS (45 degrees to straight grain), no added allowance.

Supply at least 1928.7 mm usable binding after joins; shown strips supply 1980.0 mm.

Join short ends with 10 mm straight seams; press open. Shorter strips need extra length for each additional join.

Fold outer long raw edges to centre; then fold in half, yielding 10 mm finished binding.

Enclose the zero-allowance garment edges; do not trim or move the marked boundary inward.

This rectangle is the RAW CUT strip; dashed quarter lines are folds, not joining seams.

MARKS ON FULL-SIZE CUTS

M2 FOLD

Cut envelope: 1001.0 x 41.0 mm. Charcoal outer cloth

The cuts.

Black outlines include allowances. Dashed blue lines show seam or finished boundaries. Diagrams on this page are not to scale.

tunic_bias_strip / Inner - bias binding strip

CUT 2 STRIPS ON TRUE BIAS; may piece shorter strips



Cut width 40 mm on TRUE BIAS (45 degrees to straight grain), no added allowance.

Supply at least 1291.6 mm usable binding after joins; shown strips supply 1980.0 mm.

Join short ends with 10 mm straight seams; press open. Shorter strips need extra length for each additional join.

Fold outer long raw edges to centre; then fold in half, yielding 10 mm finished binding.

Enclose the zero-allowance garment edges; do not trim or move the marked boundary inward.

This rectangle is the RAW CUT strip; dashed quarter lines are folds, not joining seams.

MARKS ON FULL-SIZE CUTS

M2 FOLD

Cut envelope: 1001.0 x 41.0 mm. Slate-blue inner cloth

tabard / Shoulder tabard

CUT 4 = two tabards, each with two fabric layers



Sew in pairs right sides together; leave a turning gap, turn and close gap.

One finished tabard drapes over each shoulder with equal front/back drop.

Mark actual shoulder and sash levels during fitting; secure tabards with a few removable tacks.

MARKS ON FULL-SIZE CUTS

M1 SHOULDER MIDPOINT / M2 SASH LEVEL / M3 SASH LEVEL

Cut envelope: 106.0 x 921.0 mm. Charcoal outer cloth

The cuts.

Black outlines include allowances. Dashed blue lines show seam or finished boundaries. Diagrams on this page are not to scale.

sash / Short fastening sash

CUT 2 = one sash with two fabric layers



Sew right sides together around the perimeter; leave a turning gap, turn and close gap.

Finished overlap is 80 mm; position low-profile hook-and-loop on opposite faces after checking fit over all layers.

Keep fastening ends short. This clothing sash is not a load-bearing device harness.

MARKS ON FULL-SIZE CUTS
M1 OVERLAP / M2 OVERLAP

Cut envelope: 771.0 x 71.0 mm. Charcoal outer cloth

Print. Prove. Cut.

Use either the supplied A4 tile set or the variable-page-size full-size PDF. SVGs retain millimeter dimensions.

01 Check the paper scale.

Print the calibration page at Actual size / 100%. Disable Fit, Shrink and printer scaling. The square must measure 50 mm in both directions and the bar must measure 100 mm. Correct scaling before printing the pattern.

02 Assemble only what you need.

A4 tiles are grouped by piece ID, row and column. Each blue frame is 190 x 250 mm, with a 10 mm overlap. Match duplicate marks and lines in that overlap; do not butt page edges. Recheck a 100 mm interval after assembly. Cut one paper template per unique shape.

03 Use the right line.

The solid black line is the fabric cutting line. Allowances are included: do not add them twice. Dashed blue is the sewing or finished-edge boundary, depending on the edge notes. Transfer notches and grain lines with removable marks.

04 Plan the fabric before buying.

Prewash and press. Arrange all copies on the usable fabric width with grain arrows parallel to the selvage. Mirrored pairs require opposite handed pieces. Cut envelopes in the CSV are planning rectangles, not an optimized fabric layout or a guaranteed yardage.

05 Prototype the movement.

Baste a toile; fasten it at the planned waist. Reach overhead and across the body, kneel, squat, sit, take a long step and turn with the hood up. Adjust restricted seams, dragging hems, sleeve clearance and hood view, then regenerate.

Build in layers.

Press after each operation. Use the exact edge allowances and finishing notes for each piece.

01 Prepare the pieces.

Cut the required copies, marking wrong sides and pairs. Transfer matching marks. Finish raw joining edges by overlock or zigzag as appropriate. Staystitch the slanting neck edges so they do not stretch during handling.

02 Join shoulders, then sleeves.

Sew front shoulders to back shoulders, right sides together. Open the body flat. Match each sleeve top midpoint to the shoulder seam and its ends to the lower armhole marks; sew at the joining allowance.

03 Close sides and underarms.

Fold the garment right sides together. Sew each sleeve underarm and corresponding body side, pivoting at the armhole corner. Check both layers meet their marks without forcing or gathering.

04 Make and attach the hood.

Join the two hood halves on the marked crown/back edges only. Finish the face edge as specified. Match hood center back to robe center back and base ends to neckline endpoints; sew the base to the neck. Clip into angular neck allowance as needed without cutting stitches.

05 Finish openings and hems.

Use each piece note for cuffs and lower hems. Join bias strip short ends with straight seams at the joining allowance, press open, then fold lengthwise into four bands and bind the specified edges. Make the undertunic in the same shoulder/sleeve/side order.

06 Fit sash and tabards last.

Sew accessory layers right sides together at the joining allowance, leave a turning gap, turn and close it. Place short sash fasteners on the toile at the actual overlap. Fit optional tabards over the shoulders and secure them at the sash or with removable snaps.

Check the seams.

Geometric checks compare generated stitching lines. A passing result is not a wear test.

Outer robe shoulders: front / back

164.22 / 164.22 mm | difference 0.000 mm | PASS

Outer robe side seams: front / back

559.05 / 559.05 mm | difference 0.000 mm | PASS

Outer robe sleeve attachment / front+back arm opening

350.00 / 350.00 mm | difference 0.000 mm | PASS

Outer robe sleeve underside pair

117.69 / 117.69 mm | difference 0.000 mm | PASS

Outer robe sleeve centre / front arm opening

175.00 / 175.00 mm | difference 0.000 mm | PASS

Inner tunic shoulders: front / back

154.28 / 154.28 mm | difference 0.000 mm | PASS

Inner tunic side seams: front / back

328.42 / 328.42 mm | difference 0.000 mm | PASS

Inner tunic sleeve attachment / front+back arm opening

330.00 / 330.00 mm | difference 0.000 mm | PASS

Inner tunic sleeve underside pair

175.64 / 175.64 mm | difference 0.000 mm | PASS

Inner tunic sleeve centre / front arm opening

165.00 / 165.00 mm | difference 0.000 mm | PASS

Hood base / complete garment neckline

428.91 / 428.91 mm | difference 0.000 mm | PASS

Hood CB-to-shoulder notch / back half-neck

60.82 / 60.82 mm | difference 0.000 mm | PASS

Hood shoulder-to-front / front neck

153.63 / 153.63 mm | difference 0.000 mm | PASS

Also inspect the paper: paired side seams and shoulder seams, sleeve midpoint marks, matching hood endpoints, a continuous cut outline, mirrored front copies and straight grain. If a new parameter combination gives an implausible shape, change the inputs and recheck the toile.

A uniform for discovery.

Design notes and sources. The geometry is an original prototype, not a traced commercial pattern.

Make it repairable.

Use ordinary woven cloth and visible, accessible seams. Add a small removable mission patch: observe, measure, build, explain. Let repairs and experiments become part of the garment's history.

Keep the device separate.

The concept terminal has no reliable physical dimensions in this folder. Design its removable carrier after measuring the real enclosure, controls and weight. The cloth pattern does not yet include a load-bearing harness.

Reference artwork

Young scout in futuristic setting.png and Futuristic scout in a vibrant world.png, in the parent project folder. These inform appearance, not body dimensions.

Construction reference

BERNINA / WeAllSew: Sewing Tutorial - Make an Easy Kimono Top. Flat sleeve assembly, pressing and print calibration.

Hood reference

Pattern Studio 101: 2 Piece Hood Pattern Making. Neck drop and front extension affect hood balance.

Printing reference

Adobe Acrobat: Print large documents. Poster tile scaling and overlap.

Cord-free hood

US CPSC: Drawstrings in Children's Upper Outerwear. Supports choosing snaps or hook-and-loop in place of neck or hood cords.

Sources reviewed 5 September 2026. Follow the generated piece notes.